



OMNIKON

NATIONAL HACKATHON 2026



01 // 10

TEAM SKILLFORGE

Members

Jositha Chowdary

Sindhuja Sai

Theme

EdTech & Skill Development

Problem Statement

Objective Evaluation of Practical Vocational Skills

Problem ID: Omni_EdTech_3

Idea: SkillProof AI

AI-assisted, evidence-linked practical skill assessment

THE SOLUTION

A smartphone-based platform where a learner records one practical task, AI converts video into structured, rubric-linked evidence

RECORD → ANALYZE → COMPARE → FLAG → TRAINER VERIFY

HOW IT WORKS

- Guided video capture with task checklist
- AI extracts observable actions, tool usage, movement patterns & safety indicators
- Observations mapped to a task-specific competency rubric
- Timestamped evidence + confidence
- Trainer reviews flagged moments and confirms/overrides
- Final verified result becomes a Digital Skill Passport

TARGET USERS

Vocational Learners • Trainers • ITIs • Training Centres • Assessment Agencies

CORE ADVANTAGE

“AI reduces repetitive observation; the trainer remains the final decision-maker.”

CURRENT GAP

- Manual assessment depends heavily on examiner judgement
- Trainers spend time reviewing complete practical performances
- Scores often have limited supporting evidence
- Maintaining consistency across centres is difficult
- Scaling assessment increases evaluator workload

SKILLPROOF AI

Current Approach	SkillProof AI
Manual observation	AI-assisted observation
Score-focused	Evidence-linked
Full video review	Flagged-moment review
Fixed/paper rubric	Digital task rubric
Difficult to scale	Cloud-based workflow

EXAMPLE

Traditional:

"Student scored 7/10."

SkillProof AI:

"Safety criterion flagged → 00:34 → Trainer reviews → Confirm / Override"

OUTCOME

More consistent • More traceable • Less repetitive review

01 — Evidence, Not Just a Score

Every flagged criterion is linked to a specific video timestamp.

02 — Task-Specific Rubric

AI maps observable evidence against predefined competency criteria.

03 — Human-in-the-Loop

AI flags → Trainer verifies → Final result

04 — Explainable Assessment

Trainer sees what was detected and where.

05 — Expandable Framework

MVP starts with one task; rubric-based architecture supports future tasks/trades.

OUR DIFFERENTIATOR

Generic Video Upload

→ stores a video

SkillProof AI

→ analyzes

→ links evidence

→ flags criteria

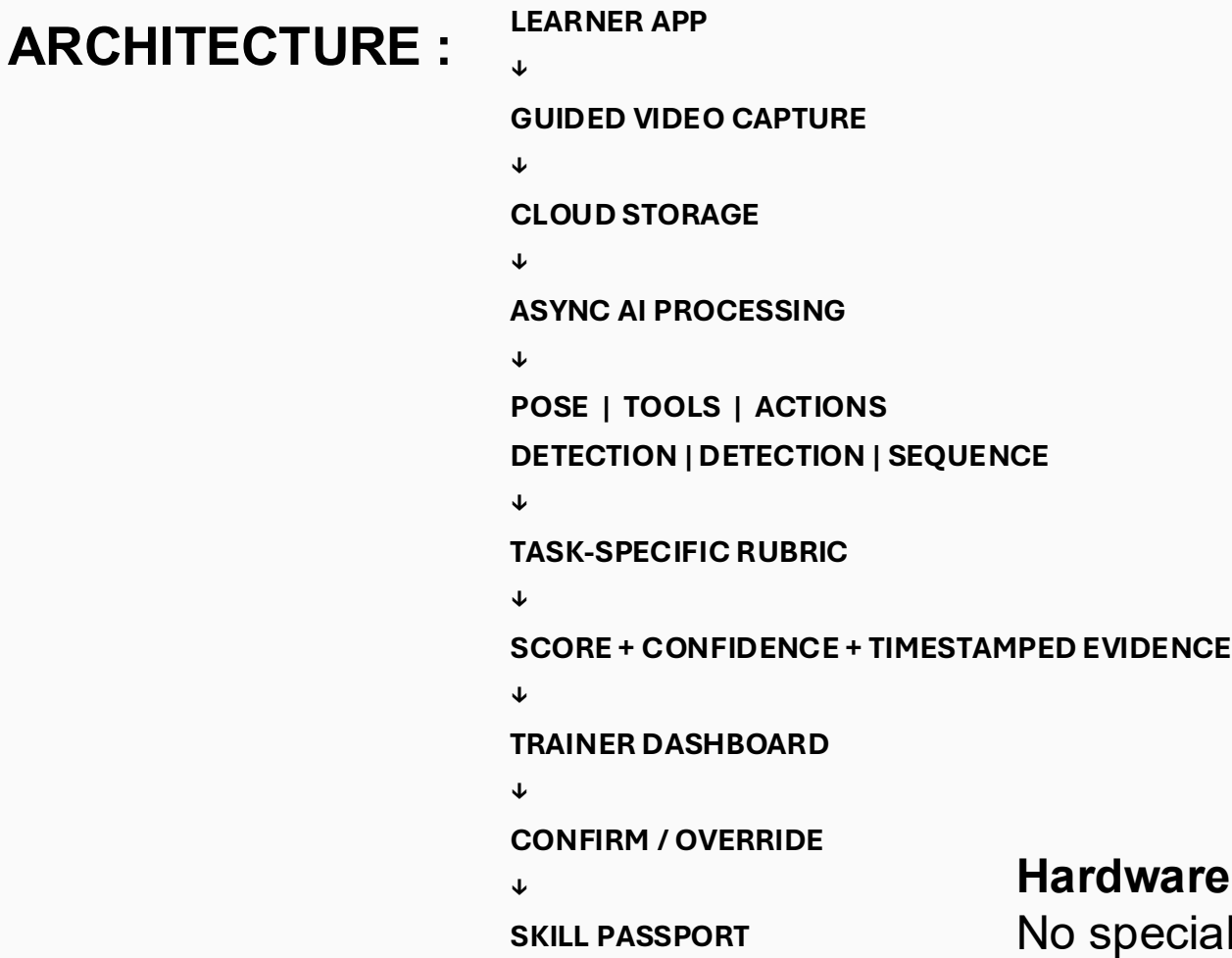
→ enables trainer verification

ONE-LINE VALUE

“Don't replace the assessor. Give the assessor better evidence.”

TECH STACK

- Frontend: React / React Native
- Backend: Python + FastAPI
- Database: PostgreSQL
- Computer Vision: MediaPipe + YOLO + OpenCV
- Processing: Python async pipeline
- Deployment: Cloud-ready



Hardware
No specialized hardware — smartphone camera only.

01 | SELECT

Choose one practical task + its rubric

02 | RECORD

Guided smartphone video capture

03 | EXTRACT

Pose • Tools • Actions • Safety cues

04 | MATCH

Compare observations with required task steps

05 | FLAG

Criterion + Evidence + Timestamp + Confidence

06 | REVIEW

Trainer checks flagged moments

07 | VERIFY

Trainer confirms/overrides findings

08 | RECORD

Generate verified Digital Skill Passport

MVP DEMO

One task → 3–5 sample videos → AI analysis → Evidence timeline → Trainer verification

FEASIBILITY & VIABILITY

07 // 10

WHY CAN WE BUILD IT?

- Pretrained CV models reduce training requirements.
- Smartphone is sufficient for video capture.
- No specialized hardware; standard smartphone + software stack.
- MVP is intentionally limited to one practical task.
- Task-specific rubrics make evaluation controllable.
- Human verification reduces risk of incorrect AI decisions.

7-DAY ROADMAP

DAY 1–2

UI + Video Capture + Backend

DAY 3–5

CV Pipeline + Task Rubric + Evidence Engine

DAY 6–7

Trainer Dashboard + Integration + Demo

MVP

MVP data: Controlled sample videos recorded by the team; real-world datasets can be incorporated later.

- ✓ Guided capture
- ✓ One practical task
- ✓ CV analysis
- ✓ Evidence flags
- ✓ Trainer verification

FUTURE

More tasks → More trades → Institute deployment → Employer verification

ONE MISSION. BUILD THE IMPOSSIBLE.

RISK → RESPONSE

Challenge	Strategy
• Lighting / camera variation	• Guided capture + confidence thresholds
• Limited task-specific data	• Start with one task + controlled sample videos
• AI misclassification	• Human verification + low-confidence escalation
• Different task requirements	• Task-specific rubrics
• Large video files	• Compression + asynchronous processing
• Student privacy	• Consent + encryption + access control

MOST IMPORTANT SAFEGUARD

AI FLAGS.

HUMAN VERIFIES.

SkillProof AI does not independently certify competency.

PRIVACY FLOW

Consent → Minimum Data → Secure Storage → Controlled Access → Deletion Policy

WHO BENEFITS?

LEARNERS

- Clearer practical feedback
- Evidence-backed skill records
- Portable Skill Passport

TRAINERS

- Less repetitive video review
- Faster access to important moments
- Structured assessment workflow

TRAINING CENTRES

- Consistent assessment process
- Centralized evidence
- Scalable evaluation

EMPLOYERS

- Better visibility into demonstrated skills
- More meaningful skill verification

MVP SUCCESS METRICS

- ↓ Review Time
- ↓ Assessment Variation
- ↑ Evidence Coverage
- ↑ Assessment Speed

Baseline vs SkillProof comparison using the same sample task.

SCALABILITY

1 TASK → 1 TRADE → MULTIPLE TRADES → INSTITUTIONAL SCALE

WHY IT MATTERS

Better evidence → Better assessment → Better trust in practical skills

RESEARCH & REFERENCES

10 // 10

PROBLEM / ASSESSMENT

[R1] NCVET — Comprehensive Revised Guidelines for Assessment Agencies

https://ncvet.gov.in/wp-content/uploads/2024/05/Comprehensive-Revised-AA-guidelines-02052024_1.pdf

[R2] NCVET — National Skills Qualification Framework

<https://ncvet.gov.in/national-skills-qualification-framework/>

[R3] NSDC — Assessment Processes & Protocols

<https://skillsip.nsdcindia.org/sites/default/files/kps-document/Assessment%20Processes%20and%20Protocols%20-%20Guide%20for%20STT%20-%20Final%2010-12-20.pdf>

TECHNOLOGY

[R4] Google MediaPipe — Pose Landmarker https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker

[R5] Ultralytics YOLO | <https://docs.ultralytics.com/>

TECHNICAL RESEARCH

[R6] ProSkill: Segment-Level Skill Assessment in Procedural Videos — WACV 2026

https://openaccess.thecvf.com/content/WACV2026/html/Mazzamuto_ProSkill_Segment-Level_Skill_Assessment_in_Procedural_Videos_WACV_2026_paper.html

REFERENCE MAPPING

- Problem & Assessment → [R1][R2][R3]
Technology → [R4][R5]
Technical Research → [R6]